

Khulna University of Engineering & Technology
B.Sc. Engineering 1st Year 1st Term (Regular) Examination, 2019
Department of Materials Science and Engineering

EEE 1127
Electrical Engineering Fundamentals

Time: 3 hours

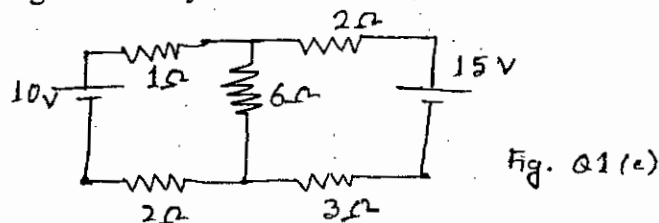
Full Marks: 210

- N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks
(iii) Assume reasonable data if any missing

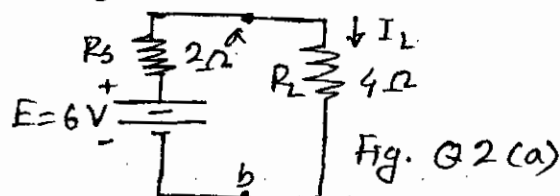
Section A

(Answer **ANY THREE** questions from this section in **answer script A**)

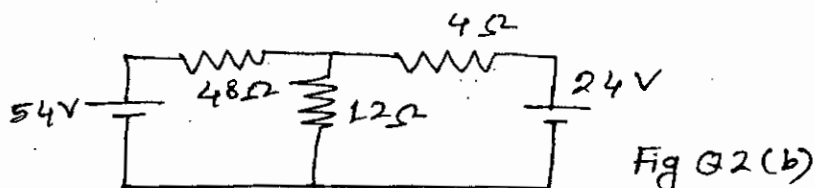
- Q 1. (a) State mathematically and graphically, Ohm's law, Kirchhoff's voltage and Kirchhoff's current law. (12)
- (b) Determine each of the 'wye' connected resistors in terms of three 'delta' connected resistors while transformation from 'delta network' to 'wye network'. (10)
- (c) Define Branch, Node and Mesh. Find the current through each branch of the network showing in **Figure-Q1(c)**, using Mesh analysis. (13)



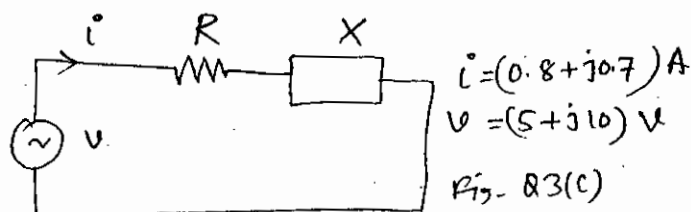
- Q 2. (a) State and explain maximum power transfer theorem. For the **Figure-Q2(a)**, (i) determine the current, I_L (ii) convert the voltage source into current source. (15)



- (b) State and explain Thevenin's and Norton's theorem. State superposition theorem. Determine the current through 4Ω resistor using this theorem for the network shown in **Figure-Q2(b)**. (20)



- Q 3. (a) Why sinusoidal wave is universally accepted? Find the impedance and amount of power absorbed by a pure L branch. (13)
- (b) What are the conditions of series and parallel resonance in a circuit? Explain with necessary phasor diagram. (10)
- (c) Define power factor, period, cycle and angular velocity. Calculate the value of R and X also indicate if X is inductive or capacitive for **Figure-Q(c)**; where $v = (5 + j10)V$, and $i = (0.8 + j0.7)A$ (12)

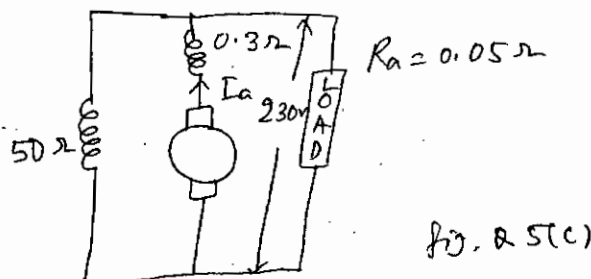


- Q 4. (a) State and explain Ampere's Circuital Law. Define magnetic field intensity and magnetic flux density. (10)
- (b) Define permeability, magnetomotive force and reluctance. Compare between electric and magnetic circuit. (10)
- (c) Define measuring instrument and classify them. How can you extend the range of ammeter and voltmeter? (08)
- (d) A moving coil instrument gives a full scale deflection of 10 mA, when the potential difference across its terminal is 100 mV. Calculate: (07)
- The shunt resistance for a full scale deflection corresponding to 100 mA.
 - Resistance for full scale reading with 1000 V.

Section B

(Answer ANY THREE questions from this section in answer script B)

- Q 5. (a) Explain briefly the working principle of DC generator. How can we get unidirectional current from a DC generator? (12)
- (b) Define critical resistance. Explain the voltage build up process for self-excited DC generator. (13)
- (c) Calculate the generated *emf* for the Figure-Q5(c). (10)



- Q 6. (a) What is the significance of back *emf* of a DC motor? Derive the condition for maximum gross mechanical power developed by a DC motor. (12)
- (b) What is the necessity of a starter? Describe a 3-point starter. Write down the relevant features of a 4-point starter over 3-point starter. (13)
- (c) What are the factors controlling motor speed? Describe any one speed control method of motor. (10)
- Q 7. (a) What will happen if the primary of a transformer is supplied with a DC supply? Show that *emf* per turn (*emf*/turn) is the same in both the primary and secondary windings. (13)
- (b) What are the losses in a transformer? If P_1 and P_2 be the iron and copper losses of a transformer on full-load, find the ratio of P_1 and P_2 such that maximum efficiency occurs at half of full load. (10)
- (c) What is meant by all day efficiency of a transformer? A 100 kVA transformer has a full load loss of 3 kW, the losses being equally divided between iron and copper. During a day, the transformer operates on full load for 3 hours, one half load for 4 hours, 75% of full load for 4 hours, the output being negligible for rest of the day. Calculate the all-day efficiency. (Assume, p.f.=1) (12)
- Q 8. (a) Derive torque equation under running condition of Induction motor. Hence, explain how rotor produce unidirectional torque? (10)
- (b) What is slip? Derive the relation between rotor speed, slip, frequency and number of stator pole. Hence, show that both rotor and stator flux rotates synchronously. (15)
- (c) Mention the advantage of having stationary armature in alternator. Differentiate between salient pole and smooth cylindrical type of rotor of alternator. (10)

Khulna University of Engineering & Technology
B.Sc. Engineering 1st Year 1st Term (Regular) Examination, 2019
Department of Materials Science and Engineering

Math 1127
Differential and Integral Calculus

Time: 3 hours

Full Marks: 210

- N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks
(iii) Assume any reasonable data if missing

Section A

(Answer **ANY THREE** questions from this section in **answer script A**)

- Q 1. (a) Define continuity of a function. Discuss the continuity and differentiability at $x = 0$ of the function. (13)

$$f(x) = \begin{cases} 1 & \text{when } x < 0 \\ 1 + \sin x & \text{when } 0 \leq x < \frac{\pi}{2} \\ 2 + (x - \frac{\pi}{2})^2 & \text{when } \frac{\pi}{2} \leq x \end{cases}$$

- (b) Differentiate, $x^{\sin^{-1} x}$ with respect to $\sin^{-1} x$. (09)

- (c) Write the indeterminate forms. Evaluate, $\lim_{x \rightarrow 2} [\frac{1}{x-2} - \frac{1}{\log(x-1)}]$ (13)

- Q 2. (a) If, $f(x) = (\frac{a+x}{b+x})^{a+b+2x}$, Show that $f'(0) = (2 \log \frac{a}{b} + \frac{b^2-a^2}{ab}) (\frac{a}{b})^{a+b}$. (10)

- (b) If, $y = \frac{1}{x^2+a^2}$, find y_n . (11)

- (c) Discuss the maxima and minima of $f(x, y) = x^3 + y^3 - 3x - 12y + 20$. (14)

- Q 3. (a) If $y = \tan^{-1} x$, then prove that, (13)

(i) $(1+x^2) y_1 = 1$

(ii) $(1+x^2) y_{n+1} + 2nx y_n + n(n-1) y_{n-1} = 0$

Find also the value of $(y_n)_0$.

- (b) State Rolle's theorem. Verify Rolle's theorem for $f(x) = x^2 + 5x - 6$ over $[-6, 1]$. (11)

- (c) Define homogeneous function. If $V = \sin^{-1}(\frac{x^2+y^2}{x+y})$ then prove that, $xV_x + yV_y = \tan V$. (11)

- Q 4. (a) Determine the equation of tangent and normal to the curve $y(x-2)(x-3) - x + 7 = 0$ at the point where it cuts the x -axis. (12)

- (b) Find the radius of curvature and centre of curvature of the parabola, $x^2 = 4ay$ at $(2a, a)$. (12)

- (c) Find all the asymptotes of $x(x-y)^2 - 3(x^2-y^2) + 8y = 0$ (11)

Section B

(Answer ANY THREE questions from this section in **answer script B**)

Q 5. (a) Calculate, $\int \frac{dx}{(x^2+1)\sqrt{x^2+4}}$ (12)

(b) Calculate, $\int \sin^{-1}\left(\sqrt{\frac{x}{x+a}}\right) dx$ (12)

(c) Calculate, $\int \frac{dx}{(2x+3)\sqrt{x^2-2x-2}}$ (11)

Q 6. (a) Obtain a reduction formula for $\int \cos^n x dx$ and hence calculate, $\int \cos^5 x dx$. (12)

(b) Evaluate, $\int_0^1 \frac{\log(1+x)}{1+x^2} dx$ (11)

(c) Evaluate, $\int_0^\infty \frac{x dx}{(1+x)(1+x^2)}$ (12)

Q 7. (a) Evaluate, $\int_0^\pi \frac{x \sin x}{1+\cos^2 x} dx$ (11)

(b) Evaluate, $\text{Lt}_{n \rightarrow \infty} \left[\frac{1}{\sqrt{2n-1^2}} + \frac{1}{\sqrt{4n-2^2}} + \frac{1}{\sqrt{6n-3^2}} + \dots + \frac{1}{n} \right]$ (10)

(c) Define Beta function and Gamma function. Find the relation between Beta and Gamma functions. (14)

Q 8. (a) Use double integral, find the area of the region enclosed between the curves, $y^2 = 4ax$ and $x^2 = 4ay$ (11)

(b) Evaluate $\int \int_R \frac{x-2y}{2x+y} dA$, where R is the region enclosed by $x-2y=1$, $x-2y=4$, $2x+y=1$ and $2x+y=3$. (14)

(c) Use triple integral, find the volume of the solid within the cylinder, $x^2 + y^2 = 9$ and between the planes $z=1$ and $x+z=5$. (10)

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Ph 1127
Optics and Waves

Time: 3 hours

Full Marks: 210

- N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks.
(iii) Assume reasonable data if any missing.

Section A

(Answer **ANY THREE** questions from this section in **answer script A**)

- Q 1. (a) What is simple harmonic motion. Show that simple harmonic motion is an oscillatory motion in terms of its displacement, velocity and acceleration. (12)
- (b) Show that the superposition principle is valid only in case of linear homogeneous equation. (13)
- (c) The force and displacement of a simple dynamic system undergoing sinusoidal excitation are given by the equations (10)

$$F = 10 \sin\left(\frac{\pi t}{10}\right) \text{ Newtons}$$

$$y = 0.10 \sin\left(\frac{\pi t}{10} - \frac{\pi}{3}\right) \text{ Meters}$$

Calculate the work done by the excitation force in (i) 10 seconds and (ii) 2.5 minutes

- Q 2. (a) Explain free, forced and damped vibration. (10)
- (b) Find an expression for a plane progressive wave and then show that, $\frac{d^2 y}{dt^2} = V^2 \left(\frac{d^2 y}{dx^2} \right)$ (15)
- (c) The displacement (in meters) of a particle executing SHM at any instant of time is given by (10)

$$y = 0.1 \sin 2\pi (340 t - 0.15)$$

Calculate (i) the amplitude of the vibrating particle, (ii) wave velocity, (iii) wave length, (iv) frequency and (v) time period.

- Q 3. (a) Show that, in the case of progressive longitudinal waves particle velocity = wave velocity $\times$ compression. (08)
- (b) Discuss Doppler's effect in sound and obtain an expression for the apparent frequency of the note when the source and the listener are (i) moving towards each other and (ii) moving away from each other. (17)
- (c) An observer on a railway platform observed a train passing through the station at a speed a . (10)
Show that the frequency of the whistle changes by, $nv \left[\frac{2a}{v^2 - a^2} \right]$; where the symbols have their usual meanings.

- Q 4. (a) What is meant by audible, ultrasonic, infrasonic and supersonic waves? Write some applications of ultrasonic waves. (12)
- (b) Show that the acoustic intensity or Intensity of sound varies directly as the square of the excess pressure (i.e. $I = \frac{P_{rms}^2}{\rho c}$). (13)
- (c) Calculate the change in intensity level when the intensity of sound increases by 10^5 times its original intensity. (10)

Section B

(Answer **ANY THREE** questions from this section in **answer script B**)

- Q 5. (a) What is visual angle? What is the necessity to know visual angle? (10)
- (b) What is compound microscope? Derive an expression for magnifying power of a compound microscope. (15)
- (c) A person has his near point at 20 cm and a range of distinct vision of 80 cm. What will be his range of vision if he wears close fitting spectacles having a power of -0.5 D? (10)
- Q 6. (a) What is coherent source of light? How can you obtain coherent source of light? (10)
- (b) What are Newton's rings and how are they formed? Describe the method for the determination of refractive index of a liquid using there rings. (15)
- (c) In a Newton's rings experiment the diameter of the 12th ring changes from 1.50 to 1.35 cm when a liquid is introduced between the lens and the glass plate. Calculate the refractive index of the liquid. (10)
- Q 7. (a) What do you mean by diffraction of light? Distinguish between Fresnel and Fraunhofer classes of diffraction. (10)
- (b) For single slit Fraunhofer diffraction, show that, $I = I_0 \frac{\sin^2 \alpha}{\alpha^2}$ (15)
- (c) Find the half angular width of the central bright maximum in the Fraunhofer diffraction pattern of slit width 10^{-4} cm when the slit is illuminated by monochromatic light of wavelength 500 nm. (10)
- Q 8. (a) What is double refraction? What is the difference between ordinary and extraordinary ray of light? (10)
- (b) Give the constructional details of a Nicol prism. How can you use it as a polarizer and an analyzer? (15)
- (c) Find the specific rotation of a given sample of sugar solution if the plane of polarization is turned through 26.4° . The length of the tube containing 20% sugar solution is 20 cm. (10)

Khulna University of Engineering & Technology
B.Sc. Engineering 1st Year 1st Term (Regular) Examination, 2019
Department of Materials Science and Engineering

Ch 1127
Inorganic and Physical Chemistry

Time: 3 hours

Full Marks: 210

- N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks
(iii) Assume reasonable data if any missing

Section A

(Answer **ANY THREE** questions from this section in **answer script A**)

- Q 1. (a) Write down the factors that affect the strength of metallic bond. (10)
(b) "Metallic bonds are non-directional"—explain with diagram. (08)
(c) By using Born Haber Cycle for the formation of AgCl, calculate the lattice enthalpy of AgCl from the following data: (07)
 Enthalpy of atomization of Ag(s): 284 kJ/mol
 1st ionization energy of Ag (g): 731 kJ/mol
 Enthalpy of atomization of Cl₂ (g): 122 kJ/mol
 1st electron affinity of Cl (g): -349 kJ/mol
 Enthalpy of formation of AgCl(s): -127 kJ/mol
(d) Discuss the formation of O₂ molecule with the help of molecular orbital theory (MOT). Why does molecule show paramagnetic property? (10)
- Q 2. (a) Write the name of different forms of corrosion that we commonly come across. Discuss reaction mechanism of pitting corrosion with diagram. (10)
(b) Explain the ways to protect or minimize corrosion. (07)
(c) What is direct loss and indirect loss due to corrosion? Explain with examples. (10)
(d) Write short note on "Environmental corrosion". (08)
- Q 3. (a) What is molar conductance? Draw the conductance titration curve of a weak acid against a strong base and explain it. (10)
(b) State and explain Faraday's law of electrolysis. (08)
(c) 0.1978 g of copper is deposited by a current of 0.2 amp in 50 minutes. What is the electrochemical equivalent of copper? (07)
(d) "Metal has high thermal and electrical conductivity"—Explain. (10)
- Q 4. (a) What are colloids? How are they classified? (06)
(b) Discuss the origin of charge on the colloidal particles. (10)
(c) What is electrodialysis? Write down the applications of colloids. (10)
(d) Explain why—? (09)
 (i) Lyophilic colloids called reversible colloids
 (ii) True solutions do not show tyndall effect
 (iii) Ionic micelles show high conductance

Section B

(Answer ANY THREE questions from this section in answer script B)

- Q 5. (a) Define energy of activation. Illustrate the transition state theory of reaction rates. How this theory considered superior to the collision theory. (10)
- (b) Discuss briefly the temperature dependence of reaction rates with the help of Arrhenius equation. (09)
- (c) What is solid state reactions? Mention the advantages of solid state reactions over solution phase reactions. (08)
- (d) "High molecularity reactions are rare"—explain with diagram. (08)
- Q 6. (a) State Raoult's law for lowering of vapor pressure. How is this law used for the determination of molecular mass of a non-volatile solute? (12)
- (b) Write down the difference between ideal and non-ideal solutions. (08)
- (c) What will be the boiling point of solution containing 0.6 g of urea (molecular mass=60) in 2.5 g of water? (Boiling point of water =373 K, $K_b=0.52$ K/mol–kg) (06)
- (d) Write short note on "steam distillation". (09)
- Q 7. (a) How many phases are present in each of the following systems: (06)
- (i) A drop of water placed in a stoppered bottle.
- (ii) A piece of molten ice placed in a beaker cover with a watch glass
- (iii) Mixture of N_2 , H_2 and O_2 .
- (b) Describe graphical and half-life method for the determination of order of reaction. (12)
- (c) Write a short note on: (10)
- (i) Steady state approximation
- (ii) Rate law of reaction approaching equilibrium
- (d) 15 mL of H_2SO_4 is dissolved in 200 mL of H_2O . Calculate the strength of the solution in molarity. [The density of H_2SO_4 and H_2O are 1.98 and 0.9987 gm per mL respectively] (07)
- Q 8. (a) Discuss the simple eutectic system of silver-lead (Ag–Pb) with a neat diagram. (10)
- (b) Explain the following term giving examples— (12)
- (i) Component (ii) Degree of freedom (iii) Congruent melting point
- (c) Derive Gibbs phase rule thermodynamically. (08)
- (d) Triple point is invariant—why? (05)

Khulna University of Engineering & Technology
B.Sc. Engineering 1st Year 1st Term (Regular) Examination, 2019
Department of Materials Science and Engineering

MSE 1101
Introduction to Materials Science and Engineering

Time: 3 hours

Full Marks: 210

- N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
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(iii) Assume reasonable data if any missing

Section A

(Answer **ANY THREE** questions from this section in **answer script A**)

- Q 1. (a) What is Materials Science and Engineering and why it is important to study this subject? (11)
Explain your answer with the view of historical perspective of evolution of this branch of engineering.
- (b) With the help of appropriate diagrams, explain the following: (14)
(i) The dependence of repulsive, attractive, and net forces on interatomic separation for two isolated atoms;
(ii) The dependence of repulsive, attractive, and net potential energies on interatomic separation for two isolated atoms.
- (c) Explain why covalently bonded materials are generally less dense than ionic or metallic bonded materials. (10)
- Q 2. (a) What is crystalline and non-crystalline material? Explain with proper example. (10)
- (b) Why are defects important in materials? Discuss the Hume-Rothery rules of solid solution formation. (15)
- (c) Calculate the fraction of atom sites that are vacant for copper (Cu) at its melting temperature of 1084°C. Assume an energy for vacancy formation of 0.90 eV/atom. The atomic weight and density for copper are 63.5 g/mol and 8.4 g/cm³ respectively at 1084 °C. (10)
- Q 3. (a) Explain the mechanism of dislocation motion through a single crystal with neat sketches. (15)
- (b) What is the difference between resilience and toughness? Describe the features of a typical stress-strain curve. (10)
- (c) Explain any **ONE** of the following strengthening mechanism of metal: (10)
(i) Solid-solution strengthening
(ii) Strain hardening
- Q 4. (a) Differentiate between ductile and brittle fracture of material. (10)
- (b) Explain the stages of ductile fracture of a material with neat sketches. (10)
- (c) What is fatigue? Write down the characteristics of fatigue. Define fatigue strength and fatigue life with the help of a fatigue curve. (15)

Section B

(Answer **ANY THREE** questions from this section in **answer script B**)

- Q 5. (a) What is meant by the “Principle of combined action” in composites? Briefly discuss about particle-reinforced composites. (15)
- (b) Explain the influence of fiber length on properties of a fiber-reinforced composite material. (10)
- (c) Is it possible to produce a continuous and oriented aramid fiber-epoxy matrix composite having longitudinal and transverse moduli of elasticity of 35 GPa and 5.17 GPa, respectively? Why or why not? Assume that the elastic modulus of the epoxy and Aramid fiber are 3.4 GPa and 131 GPa, respectively. Given, $V_m + V_f = 1$; $\sigma_c = \sigma_m V_m + \sigma_f V_f$; $F_c = F_m + F_f$; $\sigma_{cl}^* = \sigma_m' V_m + \sigma_f' V_f$; $E_{cl} = E_m V_m + E_f V_f$; $E_{ct} = \frac{E_m E_f}{E_m V_f + E_f V_m}$ symbols have their usual meaning. (10)
- Q 6. (a) In terms of electron energy band structure, discuss reasons for the difference in electrical conductivity among metals, semiconductors, and insulators. (15)
- (b) Explain the temperature dependence of carrier concentration of an intrinsic and extrinsic semiconductor. (10)
- (c) For intrinsic gallium arsenide, the room temperature electrical conductivity is $3 \times 10^{-7} (\Omega \cdot m)^{-1}$. The electron and hole mobilities are, respectively, 0.80 and 0.04 m²/V.s. Compute the intrinsic carrier concentration at room temperature. (10)
- Q 7. (a) Explain thermal expansion using the potential energy Vs. interatomic spacing curve. (10)
- (b) What are phonons? Metals have a higher thermal conductivity than ceramics – why? (10)
- (c) Briefly explain why metals are opaque to electromagnetic radiation having photon energies within the visible region of the spectrum. (07)
- (d) Explain the working principle of LASERS. (8)
- Q 8. (a) Write few common properties of ceramics. Briefly explain at what condition it is advantageous to use ceramics over metallic materials. (15)
- (b) Explain the origins of magnetic moment in a material. (10)
- (c) Describe the formation of hysteresis loop in a ferromagnetic material. (10)